WASP-35b

R-1617

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-35_171125` · created 2026-08-02T04:02:11+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458082.86329 +/- 0.00074 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1306 +/- 0.0056
Transit depth (Rp/Rs)^2	1.7 +/- 0.15 [%]
Orbital Inclination (inc)	89.88 +/- 0.62
Airmass coefficient 1 (a1)	1.013 +/- 0.0062
Airmass coefficient 2 (a2)	-0.0084 +/- 0.0033
Scatter in the residuals of the lightcurve fit is	0.62 %
Best Comparison Star	#2 - [336, 216]
Optimal Aperture	4.26
Optimal Annulus	17.61
Transit Duration (day)	0.13641 +/- 0.00091

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1306 ± 0.0056 vs 0.1241 (deviation 1.16σ)
Duration fit vs published	0.13641 d vs 0.1317 d (deviation 5.21σ)
Mid-time O–C	-3.4 min
Depth SNR	23.3
Residual scatter	0.62 %

Light curve

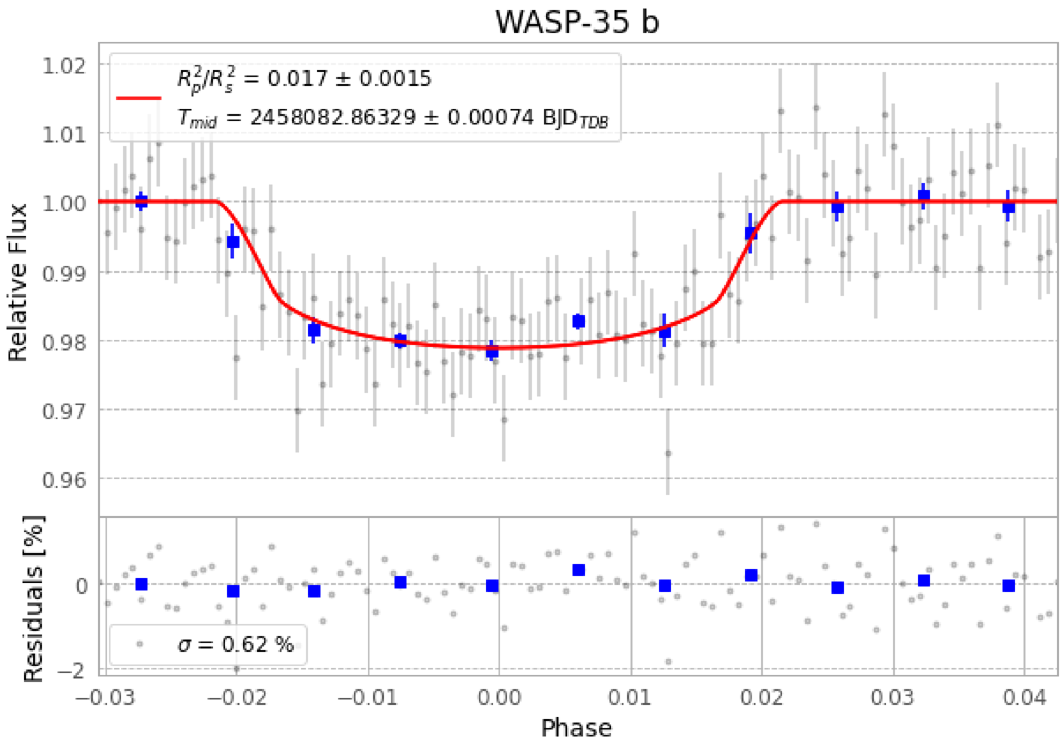

Figure 1 — FinalLightCurve_WASP-35 b_2017-11-24.png (SHA-256 101d79fcd11a050e…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [253, 249], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 ce1f63b3183c295d…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

112 FITS frames (74 MB), `WASP-35171125061513.FITS` → `WASP-35171125114812.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-171125.log (349b51d1ddf3…), FinalLightCurve_WASP-35 b_2017-11-24.png (101d79fcd11a…), FinalLightCurve_WASP-35 b_2017-11-24.csv (fb57067abdf8…)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-08-02 04:02Z.